FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2025
FULL TEST – I
PAPER –1
TEST DATE: 26-12-2024

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. C

Sol. $\frac{a}{u_x}+\frac{a}{eu_x}+\frac{a}{e^2u_x}=\frac{2u_y}{g}$ …(1)

Horizontal range:

$R=\frac{2.u_xu_y}{g}$ …(2)

2. B

Sol. $N_1=\mu N_2$ …(1)

$W=\mu N_1+N_2$ …(2)

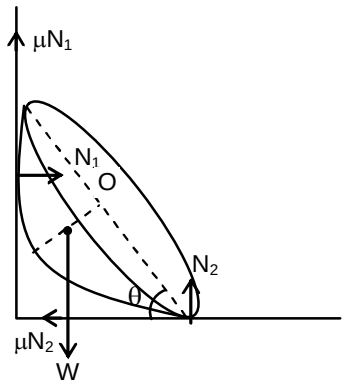
By (1) and (2)

$N_1=\frac{\mu W}{1+\mu^2}$ …(2)

Balancing torque about 'O'

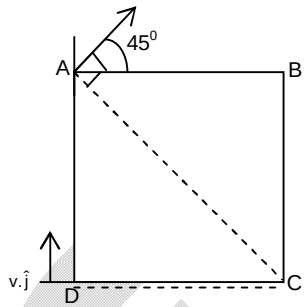
$\mu N_1\times R+\mu N_2\times R=W\times\frac{3R}{8}\sin\theta$

$\text{Sin}\,\theta=\frac{8}{3}\left[\frac{\mu+\mu^2}{1+\mu^2}\right]$



3. D

Sol. $v\left(\hat{i}+\hat{j}\right)$



The motion can be considered as rotation about an instantaneous centre of rotation. To get it, draw normals to the velocities at A and D.

From the construction, C is the center of rotation so that $V_C = 0$

$$\frac{V_C}{V_B} = 0$$

4. C

Sol. $v^2 = v_0^2 - 2gR(1-\cos\theta)$

$$v^2 = 7gR + 2gR\cos\theta$$

$$T - mg\cos\theta = \frac{mv^2}{R}$$

$$T = 7mg + 3mg\cos\theta$$

$$v_h = \sqrt{7gr} \Rightarrow T = 7mg$$

5. ABCD

Sol. Apply ampere's law

$$\oint \vec{B}\cdot d\vec{\ell} = \mu_0 i$$

6. ABC

Sol. Loop ABEF

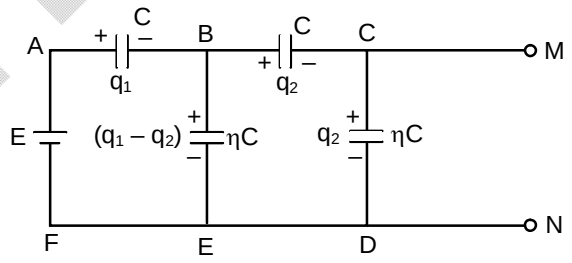
$$\frac{q_1}{c} + \frac{q_1 - q_2}{\eta c} = E \quad \ldots(1)$$

Loop ABCDEF

$$\frac{q_1}{c} + \frac{q_2}{c} + \frac{q_2}{\eta c} = E \quad \ldots(2)$$

Bu (1) and (2) we get

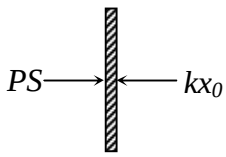
$$V_{MN} = \frac{q_2}{\eta c} = \frac{E}{\eta^2 + 3\eta + 1}$$



7. ABCD

Sol. Equilibrium of piston gives

$PS = kx_0$ or $P = \frac{kx_0}{S}$



Since, the chamber is thermally insulated $\Delta Q = 0$

$\therefore$ Elastic potential energy of spring = work done by gas

or work done by gas = $\frac{1}{2}kx_0^2$

This work is done in the expense of internal energy of the gas.

Therefore, internal energy of the gas is decreased by $\frac{1}{2}kx_0^2$.

Internal energy of an ideal gas depends on its temperature only. Internal energy of the gas is decreasing. Therefore, temperature of the gas will decrease.

8. B

Sol. Just after cutting force of cut spring will be zero whereas the force of other spring will be unchanged.

9. A

Sol. (P) $\frac{mg}{2}\frac{\ell}{4} = T\frac{3\ell}{4} \Rightarrow T = \frac{mg}{6}$

(Q) $f_r = mg\frac{\sqrt{3}}{2}$

(R) $N = \frac{mg}{2} \times \frac{4}{7} = \frac{2mg}{7}$

(S) $N = \frac{mg}{2} - \frac{mg}{6} = \frac{mg}{3}$

10. B

Sol. From the graph, first velocity increases then decreases and further change its direction (DE).

11. C

Sol. $\oint \bar{B} \cdot \overline{d\ell} = -A\left(\frac{dB}{dt}\right)$

SECTION – B

12. 8

Sol. $\frac{4\ell_1}{3} = \frac{2\ell_2}{4}$

13. 2

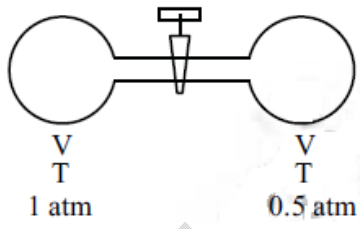
Sol. For pulse $\frac{dx}{dt} = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{3-10kt}{\mu}}$

$\Rightarrow \int_0^L dx = \int_0^{t_0} \sqrt{\frac{3-10\,kt}{\mu}}\, dt$

And $3 - 10\,k\,t_o = 0$

14. 6

Sol. After opening of at equilibrium temperature and pressure of whole gas is T_1 and P_1



$n_1 = \frac{1\times V}{RT}$, $n_2 = \frac{0.5\times V\times 4}{RT}$

$n_1 + n_2 = n$

$\frac{V}{RT} + \frac{V\times 4}{2RT} = \frac{5VP_1}{RT_1}$

$\frac{3V}{RT} = \frac{5VP_1}{RT_1}$; $\frac{P_1}{T_1} = \frac{0.6}{T}$

$\Delta Q = 0$, $\Delta W = 0$

$\therefore\ \Delta U = 0$

$n_1C_VT + n_2C_VT = (n_1 + n_2)C_VT_1$

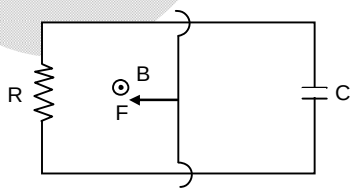
$T_1 = T$

$\frac{P_1}{T} = \frac{0.6}{T}$

$P_1 = 0.60$ atm $\Rightarrow$ $10P = 6$.

15. 4

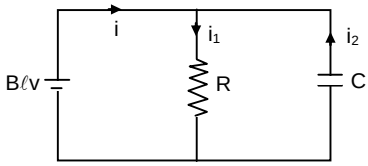
Sol. $B\ell v = i_1R$ …(1)



$B\ell v = \frac{q}{c}$ but $\frac{dq}{dt} = i_2$

$q = B\ell vc$

$i_i = B\ell ca$ …(2)



$F - (i_1 + i_2)\, B\ell = ma$

Solving this we can find terminal velocity

$v = \frac{FR}{B^2\ell^2} = \frac{FR}{4FR} = 0.25$; $\frac{1}{v} = 4$.

16. 2

Sol. Use formula in terms of reduced mass

$$\frac{1}{\lambda} = \frac{R_\infty z^2}{\left(1+\frac{m}{M}\right)}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$\therefore\ \frac{\lambda_D}{\lambda_H} = \frac{1+\frac{m}{M_D}}{1+\frac{m}{M_H}}$$

$\frac{M_D}{M_H} \approx 2.0$

17. 7

Sol. For process AB $T_A = 300\ K,\ T_B = 600\ K$

$W = nR\Delta T = nR(T_B - T_A) = 300\ nR\ = 600R.$

$Q = n\ Cp\ \Delta T = 2 \times \frac{5}{2} R\ (300) = 1500\,R.$

For process BC $W = nRT \ln \frac{v_f}{v_c} = nRT \ln \frac{p_i}{p_f}\ = nRT \ln 2\ = 1200R \ln 2$

$Q = W = 1200R \ln 2$

For process CA $W = \int P\ dV = \int_{600}^{300} \frac{K}{T} \frac{2nRT}{K} dT.$

$= -2nR(300) = -1200R.$

$Q = nC_V\ \Delta\ T + W$

$= 2 \times \frac{3}{2} R(-300) - 1200R.$

$= -900R - 1200R = -2100R$

$\eta = \frac{600R + 1200R \ln 2 - 1200R}{1500R + 1200R \ln 2}$

$= 1 - \frac{21}{12 \ln 2 + 15} \Rightarrow x = 7.$

Chemistry

PART – II

SECTION – A

18. B

Sol. $P_4O_{10} + 4HNO_3 \longrightarrow 2N_2O_5 + 4HPO_3$

19. B

Sol. Each photon strikes out one electron form the metal surface. Intensity will increase the number of photons, so number of photoelectrons increase.

20. C

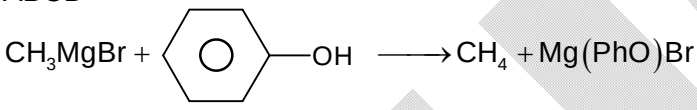
Sol. In(C), the complex ions have zero charge. So there is no attraction between the two.

21. B

Sol. $\alpha = \frac{\Lambda_m}{\Lambda_m^0}$

22. ABCD

Sol.



$CH_3MgBr +$ —OH $\longrightarrow CH_4 + Mg(PhO)Br$

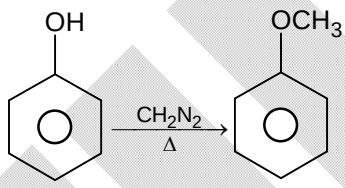
23. AB

Sol. $PH_4Cl \xrightarrow{\Delta} PH_3 + HCl$

$4H_3PO_3 \xrightarrow{\Delta} 3H_3PO_4 + PH_3$

24. ABD

Sol.



25. B

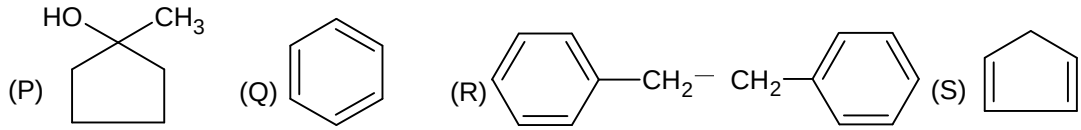
Sol. CsCl has b.c.c unit cell, which coordination number equal to eight.

26. D

Sol. Due to half-filled $t_{2g}^3 e_g^2$ configuration $[FeF_6]^{3-}$ complex is colourless.

27. C

Sol. The products are



28. C

Sol. $Fe^{3+} \rightarrow Fe(SCN)_3$

$Fe^{2+} \rightarrow$ Green sulphate

$Fe^{4+}(n = 4)$, $Fe^{2+}(n = 4)$

$Fe \rightarrow Fe(CO)_5$ hybridization(dsp^3)

SECTION – B

29. 864

Sol. $P_{equilibrium} = p_B + 2p_C = 18$ atm

Since $p_B + 2p_C = 3x = 18$

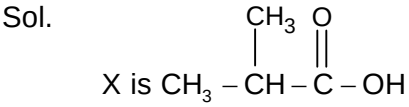
or $x = 6$

$\therefore p_B = 6$ atm, $p_C = 12$ atm

$\therefore K_p = p_B \times (2p_C)^2 = 6 \times (12)^2 = 864$

30. 88

Sol.



X is $CH_3 - CH(CH_3) - C(=O) - OH$

31. 801

Sol. $\mu_{max} = q \times d = 1.602 \times 10^{-19} \times 5 \times 10^{-10} = 8.01 \times 10^{-29}$ coulomb meter = b x 10^{-29}

$\therefore$ 100 b = 801

32. 120

Sol. X is mesitylene.

Y is Chloro mestiylene

33. 80

Sol. $A = Na_2SO_4.10H_2O$, $B = Na_2SiO_3$, $C = SO_3$, D = HCl, PPt = $BaSO_4$

34. 5

Sol. Moles of $Sn^{2+} = 523.15 \times 10^{-3}$

Moles of MnO_4^- required = $\frac{2}{5} \times 523.15 \times 10^{-3}$

$$\frac{\frac{2}{5} \times 523.15 \times 10^{-3} \times 1000}{418.52} = x$$

$x = 0.5$, $10x = 5$

Mathematics

PART – III

SECTION – A

35. B

Sol. $f(x)=2\operatorname{cosec}2x+\left(\sqrt{\sec x}-\sqrt{\operatorname{cosec}x}\right)^2+\dfrac{2\sqrt{2}}{\sqrt{2\cos x.\sin x}}$

$\therefore\ f(x)=\dfrac{2}{\sin 2x}+\left(\sqrt{\sec x}-\sqrt{\operatorname{cosec}x}\right)^2+\dfrac{2\sqrt{2}}{\sqrt{\sin 2x}}$

So, $f_{min}\left(x=\dfrac{\pi}{4}\right)=\dfrac{2}{1}+0+\dfrac{2\sqrt{2}}{\sqrt{1}}=2\left(\sqrt{2}+1\right)$

36. D

Sol. $f(x)=\tan^{-1}\left(2x^2\right)$ and $g(x)=\left(-x^2\right)$

$\Rightarrow \lim\limits_{x\to 0}\dfrac{e^{2x^2}-e^{\tan^{-1}2x^2}}{x^6}=\dfrac{8}{3}$

37. C

Sol. We have $f(x)=[3x]+14+\left|(2x+1)(2x-1)\right|(2x-1)(x+2)+\sin\left(\dfrac{\pi x}{2}\right)$

For $x\in[-4,4]$

$-12\le 3x\le 12$

$\therefore\ x=\underbrace{\dfrac{-12}{3},\dfrac{-11}{3},\dfrac{-10}{3},.....,\dfrac{-1}{3},0,\dfrac{1}{3},.....,\dfrac{11}{3},\dfrac{12}{3}}_{25\text{ points}}$

But $f(-4)=f\left(-4^+\right)$

$\Rightarrow f(x)$ is continuous at $x=-4$.

Here, $f(x)$ is continuous at $x=\dfrac{-1}{2}$ but non – derivable at $x=\dfrac{-1}{2}$.

As discontinuing $\Rightarrow$ non – differentiability

So, $f(x)$ is non – derivable at 25 points in $[-4,4]$

38. C

Sol. Since $P(1)=1+b+c$ and $P(2)=4+2b+c$ are both roots of P, we have, from Viet's that P (1) + P (2) = – b $\Rightarrow 4b+2c=-5$ and $(1+b+c).(4+2b+c)=c$ substituting $b=-\dfrac{2c+5}{4}$, we get $\left(-\dfrac{1}{4}+\dfrac{c}{2}\right).\left(\dfrac{3}{2}\right)=c\Rightarrow\dfrac{4c-2}{8}.\dfrac{3}{2}=c\Rightarrow c=-\boxed{\dfrac{3}{2}}$

39. BD

Sol. $L=\lim\limits_{x\to 0}\dfrac{\sin\left(x^2\right)+2\cos(bx)-ax^4-2}{e^{ax}-1-ax-2x^2-\dfrac{a^3x^3}{6}}$

$=\lim\limits_{x\to 0}\dfrac{x^2-\dfrac{x^6}{3!}+....+2\left(1-\dfrac{b^2x^2}{2!}+\dfrac{b^4x^4}{4!}+....\right)-ax^4-2}{1+ax+\dfrac{a^2x^2}{2!}+\dfrac{a^3x^3}{3!}+\dfrac{a^4x^4}{4!}+....-1-ax-2x^3-\dfrac{a^3x^3}{6}}$

$=\lim\limits_{x\to 0}\dfrac{\left(1-b^2\right)x^2+\left(\dfrac{b^2}{12}-a\right)x^4+.....}{\left(\dfrac{a^2}{2}-2\right)x^2+\dfrac{a^4x^4}{24}+....}$

$\dfrac{1-b^2}{\dfrac{a^2}{2}-2}=\dfrac{25}{8}\Rightarrow 16-16b^2=25\left(a^2-4\right)$

$\Rightarrow\ 25a^2+16b^2=116$

$\Rightarrow \left.\begin{matrix}a^2=4 \text{ and } b^2=1\\ a=\pm 2,\ b=\pm 1\end{matrix}\right\}\left(\because a,b\in I\right)$

$\dfrac{1-b^2}{\dfrac{a^2}{2}-2}=\dfrac{-23}{8}\Rightarrow 16-16b^2=-23\left(a^2-4\right)$

$23a^2-16b^2=76\Rightarrow a^2=4$

$\Rightarrow \left.\begin{matrix}a^2=4 \text{ and } b^2=1\\ a=\pm 2,\ b=\pm 1\end{matrix}\right\}\left(\because a,b\in I\right)$

Now, $L=\dfrac{\dfrac{b^2}{12}-a}{\dfrac{a^4}{24}}=\dfrac{2\left(b^2-12a\right)}{a^4}$

For $a=2,\ b=\pm 1$

$\Rightarrow\ L=\dfrac{2.(1-24)}{16}=\dfrac{-23}{8}$

For $a=-2,\ b=\pm 1$

$\Rightarrow\ L=\dfrac{2.(1+24)}{16}=\dfrac{25}{8}$

40. ABD

Sol. $f'(0)=0;\ f'(x)=\begin{cases}2x\sin\dfrac{1}{x}-\cos\dfrac{1}{x}, & \text{for } 0<x\le 1\\ 0, & \text{if } x=0\end{cases}$

$\Rightarrow\ f(x)$ is differentiable in $[0,1]$

$$\lim_{x\to 0}\frac{f(x)}{g(x)}=\lim_{x\to 0}\frac{x^2\sin\left(\frac{1}{x}\right)}{x^2}=\text{D.N.E.}$$

$$\lim_{x\to 0}f'(x)g'(x)=\lim_{x\to 0}\left(4x^2\sin\frac{1}{x}-2x\cos\frac{1}{x}\right)=0$$

Also, $\lim_{x\to 0}\frac{f'(x)}{g'(x)}$ does not exist.

41. ABCD

Sol. (A) Let $g(t)=e^{\frac{t^2}{2}}.f(t)$ for $t\in[a,b]$ here $g(a)=g(b)=0$

According to Rolle's theorem

$g'(x)=0$, for some $x\in(a,b)$

$\Rightarrow xe^{\frac{x^2}{2}}.f(x)+e^{\frac{x^2}{2}}.f'(x)=0$

$\Rightarrow xf(x)+f'(x)=0$

(B) Let $g(t)=e^{\frac{-t^2}{2}},f(t),$ for $t\in[a,b]$

here $g(a)=g(b)=0$

$\therefore$ According to Rolle's theorem

$g'(x)=0$ for some $x\in(a,b)$

$\Rightarrow -xe^{\frac{x^2}{2}}.f(x)+e^{\frac{-x^2}{2}}.f(x)=0$

$\Rightarrow xf(x)-f'(x)=0$

(C) Let $g(t)=e^{50t}f(t),\ t\in[a,b]$

Here $g(a)=g(b)=0$

$\therefore$ According to Rolle's theorem

$g'(x)=0$ for some $x\in(a,b)$

$\Rightarrow 50e^{50x}f(x)+e^{50x}f'(x)=0\Rightarrow 50f(x)+f'(x)=0$

(D) Let $g(t)=50^t f(t),\ t\in[a,b]$

Here $g(a)=g(b)=0$

$\therefore$ According to Rolle's theorem

$g'(x)=0$ for some $x\in(a,b)$

$\Rightarrow 50^x\ln 50f(x)+50^x f'(x)=0$

$\Rightarrow(\ln 50)f(x)+f'(x)=0$

42. B

Sol. (P) $A^2+B=I$ ....(1)

$BA^2+B^2=B$

$\Rightarrow B^2 = B$

$\therefore A^2 + B^2 = 1 \Rightarrow \text{tr}\left(A^2 + B^2\right) = 3$

(Q) $x\dfrac{dy}{dx} + y(\ln y) = 0 \Rightarrow \int \dfrac{dx}{x} + \int \dfrac{dy}{y(\ln y)} = C;$

$\ln(x \ln y) = C$. If $x = 1$ then $y = e$

Hence, $\Rightarrow \ln(\ln e) = c \Rightarrow c = 0$

$\Rightarrow \ln(x \ln y) = 0 \Rightarrow x \ln y = 1 \Rightarrow y = f(x) = e^{\frac{1}{x}}$

Now, $\lim\limits_{x\to\infty} x\left(f(x) - 1\right) = \lim\limits_{x\to\infty}\left(e^{\frac{1}{x}} - 1\right) = 1 = e^0$

$\Rightarrow \lambda = 0]$

(R) $\int_1^e \left(2^{\ln x} + e\right)dx + \int_{1+e}^{2+e} e^{\log_2(x-e)}\,dx$

$= e(2 + e) - (1 + e)$

$\Rightarrow \int_1^e 2^{\ln x}dx + \int_{1+e}^{2+e} e^{\log_2(x-e)}\,dx$

$= (2e - 1)$

$\therefore [\lambda] = 4$

(S) As two given lines are parallel, so we must have

$(\alpha + 1 + 2(\alpha - 3) - 1)(2(\alpha + 1) + 4(\alpha - 3) - 14) < 0$

$\Rightarrow (3\alpha - 6)(6\alpha - 24) < 0 \Rightarrow 2 < \alpha < 4$

Hence only one integral value of $\alpha = 3$ exist.

Note : $P(\alpha + 1, \alpha - 3)$ lies on line $x - y = 4$

43. C

Sol. Properties of modulus.

44. A

Sol. (P) x b b b x c x c x c x

Number of ways $= \dfrac{4!}{3!} \times {}^5C_4 = 20$

(Q) 2b, 1b; 2c, 1c or 2b, 1b ; 1c, 1c, 1c or 2b, 1b ; 3c

x bb x c x c x b x c x, bbcbcc ; bcbbcc, bccbbc (same way starting with c)

cbbcbc, cbcbbc

bbcccb, bcccbb

number of ways $= 12 \times {}^6C_4 = 180$

(R) bcbcbc $= {}^7C_4$

bccbcb or bcbccb $= 2 \times {}^6C_3$

bbccbc or bccbbc or bbcbcc or bcbbcc $= 4 \times {}^5C_2$

bcccbb or bbcccb $= 2 \times {}^4C_1$

bbbccc $= {}^4C_4$

Total ways $= 2(35 + 40 + 40 + 8 + 1) = 248$

(S) bcbcbc, cbcbcb
bccbcb, cbbcbc
bcbccb, cbcbbc
number of ways $= {}^7C_1 \times 2 + 4 = 18$

45. D

Sol. (P) ${}^nC_3\left(\frac{x}{a}\right)^{n-3}\left(\frac{1}{x}\right)^3 = \frac{5}{2} \quad \Rightarrow \quad n-6=0$

$\Rightarrow \quad a^3 = 8 \quad \Rightarrow \quad a = 2$

(Q) $\sum_{p=1}^{4}\sum_{r=p}^{4}\frac{4!}{r!(4-r)!(r-p)!p!} = \sum_{p=1}^{4}{}^4C_p\sum_{r=p}^{4}\frac{(4-p)!}{(r-p)!(4-r)!}$

$= \sum_{p=1}^{4}{}^4C_p\,2^{4-p} = 3^4 - 2^4 = 65$

(R) Coefficient of x^{13} in $(1-x)\left(1-x^4\right)^4 = -\left(-{}^4C_3\right) = 4$

(S) $\sum_{r=0}^{4}{}^4C_r\,(r-2)^2 = 16$

SECTION – B

46. 2

Sol. $S_n = \sum_{r=1}^{n}\tan^{-1}\left(\frac{2(2r-1)}{4+r^2\left(r^2-2r+1\right)}\right)$

$= \sum_{r=1}^{n}\tan^{-1}\left(\frac{2(2r-1)}{4+r^2(r-1)^2}\right)$

$= \sum_{r=1}^{n}\tan^{-1}\left(\frac{\frac{2r-1}{2}}{1+\frac{r^2}{2}\cdot\frac{(r-1)^2}{2}}\right)$

$= \sum_{r=1}^{n}\tan^{-1}\left(\frac{\frac{r^2}{2}-\frac{(r-1)^2}{2}}{1+\frac{r^2}{2}\cdot\frac{(r-1)^2}{2}}\right)$

$= \sum_{r=1}^{n}\left(\tan^{-1}\frac{r^2}{2}-\tan^{-1}\frac{(r-1)^2}{2}\right)$

$S_n = \tan^{-1}\frac{n^2}{2} \Rightarrow \cot(S_n) = \frac{2}{n^2}$

$S_{n-1} = \tan^{-1}\left(\frac{(n-1)^2}{2}\right) \Rightarrow \cot\left(S_{n-1}\right) = \frac{2}{(n-1)^2}$

$$\cot\left(S_{n-1}\right)-\cot\left(S_n\right)=\frac{2}{(n-1)^2}-\frac{2}{n^2}$$

$$\lim_{n\to\infty}\sum_{n=2}^{n}\left(\cot\left(S_n\right)-\cot S_n\right)=\lim_{n\to\infty}\sum_{n=2}^{n}\left(\frac{2}{(n-1)^2}-\frac{2}{n^2}\right)$$

$$=\lim_{n\to\infty}\left[\left(\frac{2}{1^2}-\frac{2}{2^2}\right)+\left(\frac{2}{2^2}-\frac{2}{3^2}\right)+\left(\frac{2}{3^2}-\frac{2}{4^2}\right)+\left(\frac{2}{(n-1)^2}-\frac{2}{n^2}\right)\right]$$

$$=\lim_{n\to\infty}\left(2-\frac{2}{n^2}\right)=2$$

47. 45

Sol. We have, $f(x)=1-x^3$ and $g_5(x)=\dfrac{5}{\frac{1}{f(x)}+\frac{1}{f(2x)}+.....+\frac{1}{f(5x)}}$

$$1-\frac{5}{\frac{1}{f(x)}+....+\frac{1}{f(5x)}}$$

Now, $\lim\limits_{x\to0}\dfrac{1-g_5(x)}{x^3}=\lim\limits_{x\to0}\dfrac{1-\frac{5}{\frac{1}{f(x)}+.....+\frac{1}{f(5x)}}}{x^3}$

$$=\lim_{x\to0}\frac{\left(\frac{1}{f(x)}-1\right)+\left(\frac{1}{f(x)}-1\right)+.....+\left(\frac{1}{f(5x)}-1\right)}{x^3\left(\frac{1}{f(x)}+\frac{1}{f(2x)}+\frac{1}{f(3x)}+\frac{1}{f(4x)}+\frac{1}{f(5x)}\right)}$$

$$=\lim_{x\to0}\frac{\frac{(1-f(x))}{f(x)}+\frac{(1-f(2x))}{f(2x)}+....+\frac{(1-f(5x))}{f(5x)}}{5x^3}$$

$$=\frac{1^3+2^3+3^3+4^3+5^3}{5}$$

$$=\frac{\left(\frac{5(5+1)}{2}\right)^2}{5}=45$$

48. 118

Sol. $\dfrac{d}{dx}\left(f^3(x)-g^3(x)\right)=3f^2(x)f'(x)-3g^2(x)g'(x)=0$

So, $f^3(10)-g^3(10)=f^3(2)-g^3(2)$

$\Rightarrow f^3(2)=(5)^3-(2)^3+(1)^3=125-8+1=126-8=118$

49. 3

Sol. The given equation can be written as $\frac{x^2}{\alpha^2}+\frac{y^2}{4}=1$

Any point on the curve is $P(\alpha\cos\theta, 2\sin\theta)$

$\therefore$ $PQ=\sqrt{\alpha^2\cos^2\theta+4(1+\sin\theta)^2}$

Let $S(\theta)=\alpha^2\cos^2\theta+4(1+\sin\theta)^2=(4+\alpha^2)+8\sin\theta+(4-\alpha^2)\sin^2\theta$

Now, $\frac{dS}{d\theta}=2\cos\theta\left[4+(4-\alpha^2)\sin\theta\right]$

$=\frac{ds}{d\theta}=0$ gives $\theta=\frac{\pi}{2},\frac{3\pi}{2}$ or $\sin\theta=\frac{4}{\alpha^2-4}$.

But $0<\alpha^2<8$ and $|\sin\theta|\le 1$, so $\sin\theta=\frac{4}{\alpha^2-4}$ is not possible.

Also, at $\theta=\frac{3\pi}{2}$, P becomes (0, –2) which is same as point Q. So PQ = 0.

Also, $\left.\frac{d^2S}{d\theta^2}\right]_{\theta=\frac{\pi}{2}}=-8\sin\theta+2(4-\alpha^2)\cos 2\theta$

$=-8+2(4-a^2)(-1)$

$=-2(8-\alpha^2)<0$ (As $0<\alpha^2<8$)

Hence $\theta=\frac{\pi}{2}$ corresponds to the maximum values of S.

So, P is (0, 2)

Clearly, reflection of P (0, 2) in the x – axis is R (0, –2).

So, least distance of R (0, –2) from the line $3x-4y+7=0$, is $\frac{|3(0)-4(-2)+7|}{\sqrt{(3)^2+(-4)^2}}=\frac{15}{5}=3$

50. 5

Sol. $P=\lim\limits_{n\to\infty}\prod\limits_{r=1}^{n}\left(\sin\left(\frac{r\pi}{4n}\right)\cos\left(\frac{r\pi}{4n}\right)\right)^{\frac{1}{n}}$

$\ln P=\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\ln\left(\frac{\sin r\pi}{n}.\frac{\cos r\pi}{n}\right)$

$\ln P=\int_0^1\ln\left(\sin\frac{\pi x}{4}.\cos\frac{\pi x}{4}\right)dx$

$\ln P=\int_0^1\ln\left(\sin\frac{\pi x}{2}\right)dx-\ln 2$

Put $\frac{\pi x}{2}=t$

$\therefore$ $\ln P=\frac{2}{\pi}\int_0^{\frac{\pi}{2}}\ell n(\sin t)\,dt-\ell n 2$

$= -2\ell n 2 = \ell n \frac{1}{4}$

$\therefore P = \frac{1}{4} \equiv \frac{a}{b} \Rightarrow a + b = 5$

51. 6

Sol. We have, $\left|x^2 - \left(7 + k^2\right)x + 7k^2\right| + \sqrt{(x-3)(x-3k+2)} = 0$

$\Rightarrow \left|(x-7)\left(x-k^2\right)\right| + \sqrt{(x-3)\left(x-(3x-2)\right)} = 0$

The above equation has a solution if equations $\left(x-7\right)\left(x-k^2\right) = 0$ and $\left(x-3\right)\left(x-\left(3k-2\right)\right) = 0$ has at least one common root.

$\therefore k^2 = 3 \Rightarrow k = \pm\sqrt{3}$

or $k^2 = 3k - 2 \Rightarrow k^2 - 3k + 2 = 0$

$\Rightarrow (k-1)(k-2) = 0 \Rightarrow k = 1$ or 2

$3k - 2 = 7 \Rightarrow k = 3$

$\therefore$ Sum of all values of k is $= \sqrt{3} - \sqrt{3} + 1 + 2 + 3 = 6$